

Priyam Jain

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Professional Summary

Computer Science student and AWS Certified Solutions Architect with hands-on experience building backend infrastructure and machine learning systems from scratch. Focused on developing custom GenAI workflows, training ML/DL models, and creating reliable backend architectures using FastAPI and Flask. Passionate about writing efficient code and building scalable, intelligent applications from the ground up.

Education

Vellore Institute of Technology

B.Tech in Computer Science & Engineering (AI & ML) | CGPA: 8.10/10

Oct 2023 – Present

Bhopal, India

Jai Academy

Class XII (CBSE)

2023

Jhansi, India

Christ The King College

Class X (ICSE)

2021

Jhansi, India

Technical Skills

- **Languages:** Python, C++, JavaScript, SQL (MySQL), HTML, CSS
- **AI/ML & GenAI:** PyTorch, LangChain, YOLOv8, Ollama, FAISS, Keras, scikit-learn
- **Frameworks & Libraries:** FastAPI, Flask, NumPy, Pandas
- **Cloud, DevOps & MLOps:** AWS, Docker, MLflow, Apache Airflow, DVC, Dagshub, Git, GitHub

Experience

Web & WordPress Development Intern

SpWeb@Dev

Feb 2024 - April 2024

(Remote) Jhansi, India

- Developed web applications and digital solutions during targeted internships, focusing on WordPress development at **spweb@dev** and full-stack web development practices at **CodeAlpha**.

Projects

RAG Research Agent

LangChain, Ollama, FAISS, Python

- Built autonomous research assistant using RAG with LangChain agents, Ollama LLMs (Llama/Qwen), and FAISS vector database for privacy-focused local deployment without cloud dependency.
- Designed custom retrieval pipeline with metadata-based ranking, source-backed citations, and real-time streaming responses, improving reliability over traditional static RAG through agent-based decision-making.

Disaster Survivor Detection System (DSDS)

YOLOv8, PyTorch, OpenCV

- Developed real-time aerial survivor detection system processing drone imagery at 74 FPS, achieving 65.3% precision and mAP50 of 0.499 using optimized YOLOv8s for detecting small objects (10×10 px) at altitudes up to 120m.
- Applied Mosaic, Mixup, and HSV augmentation techniques to enhance model robustness, reducing manual analysis delays in critical disaster search-and-rescue operations.

Vendora – AI-Powered Service Marketplace

FastAPI, Flask, AWS, Docker, PyTorch

- Containerized and deployed a microservices architecture (Flask/FastAPI) on AWS EC2 with Application Load Balancers and Auto Scaling Groups for fault-tolerant scalability.
- Engineered a relational schema (RDS MySQL) implementing a complete booking lifecycle with conflict handling alongside a personalized vendor recommendation engine.
- Integrated a fine-tuned DistilBERT model for sentiment analysis and established CloudWatch/SNS monitoring to ensure automated infrastructure scaling and high uptime.

Certifications

- **AWS Certified Solutions Architect – Associate (SAA-C03)**
- **Deep Learning** – Smart Bridge, Artificial Intelligence

Extra-Curriculars

Leadership & Responsibilities:

- **Technical & Research Lead (Microsoft Club) & Co-Lead (MERN Matrix Club)**

Led cross-functional teams across website development, creative event planning, video editing, and technical judging. Orchestrated multi-channel digital outreach initiatives—including content writing, technical blogging, email communications, and LinkedIn campaigns—to accelerate community engagement.